Techno-Economic Assessment of Tailings Valorization at Eagle Mine, Michigan

Metal Recovery, CO_2 Mineralization, and
Sulfuric Acid Production from Ni–Cu Mine Tailings

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`Separation-Agents Framework`

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Abstract

This report presents a simulation-driven techno-economic assessment of a tailings valorization process for the Eagle Mine, a nickel–copper sulfide operation in Michigan's Upper Peninsula (Lundin Mining). The proposed process recovers residual Cu and Ni metals via acid leaching and solvent extraction, sequesters atmospheric CO_2 through mineral carbonation of magnesium silicates, and achieves sulfuric acid self-sufficiency from pyrrhotite roasting.

Process parameters are derived from three modeling tiers: (i) **Reaktoro** Gibbs energy minimization for leach pH, CO_2 carbonation, and Mg speciation; (ii) **IDAES-PSE** sequential-modular solver for separator performance; and (iii) **McCabe–Thiele** mass-balance models for Cu/Ni solvent extraction, with dissolution kinetics informed by the Reaktoro-derived leach pH. **BoTorch** Bayesian optimization is used to identify optimal process conditions.

At a throughput of 2,000 t/d, the simulation yields a leach pH of 1.53 (Reaktoro), CO_2 conversion of 99.99% (Reaktoro GEM), and overall Cu/Ni recoveries of 87% (McCabe–Thiele at pH 1.53). The resulting economics show a total CAPEX of \$9.0 M, an annualized cost of \$13.2 M/yr, and revenues of \$125.0 M/yr, yielding a **net value of \$169/t tailings**.

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1 Introduction and Opportunity

The Eagle Mine is an underground nickel–copper sulfide mine located in Marquette County, Michigan, operated by Lundin Mining Corporation. The ore body is a magmatic Ni–Cu–PGE sulfide deposit hosted in mafic and ultramafic intrusive rocks. Primary processing at Eagle Mine concentrates pentlandite (Ni) and chalcopyrite (Cu) via froth flotation, producing a bulk sulfide concentrate shipped to smelters.

The flotation tailings retain significant residual value:

(i) **Residual metals:** Approximately 0.7 wt% Ni and 0.9 wt% Cu remain locked in fine sulfide grains.

(ii) **Magnesium silicates:** Forsterite, enstatite, and lizardite provide reactive feedstock for CO_2 mineralization.

(iii) **Iron sulfides:** Pyrrhotite and magnetite can be recovered for iron production and H_2SO_4 generation.

This study develops an integrated valorization flowsheet that simultaneously addresses all three value streams, using rigorous simulation to derive process parameters rather than relying on engineering assumptions alone.

2 Tailings Composition

Table 1: Nominal mineralogical composition of Eagle Mine tailings (mole fraction basis).

Mineral	**Formula**	**Mol %**	**Role in Process**
Forsterite	Mg_2SiO_4	45	CO_2 mineralization feedstock
Enstatite	$MgSiO_3$	15	CO_2 mineralization feedstock
Pyrrhotite	$Fe_{1-x}S$	10	H_2SO_4 production
Anorthite	$CaAl_2Si_2O_8$	15	Gangue (reports to residue)
Lizardite	$Mg_3Si_2O_5(OH)_4$	10	CO_2 mineralization feedstock
Magnetite	Fe_3O_4	5	Iron concentrate product

Table 2: Residual metal grades in Eagle Mine tailings.

Metal	**Grade (wt%)**	**Hosting Mineral**
Nickel (Ni)	0.7	Fine pentlandite / violarite inclusions
Copper (Cu)	0.9	Fine chalcopyrite / bornite inclusions

The Mg-silicate content (forsterite + enstatite + lizardite) represents 70 mol% of the tailings with an estimated MgO equivalent of approximately 30 wt%.

3 Process Description

The valorization process consists of six sequential unit operations.

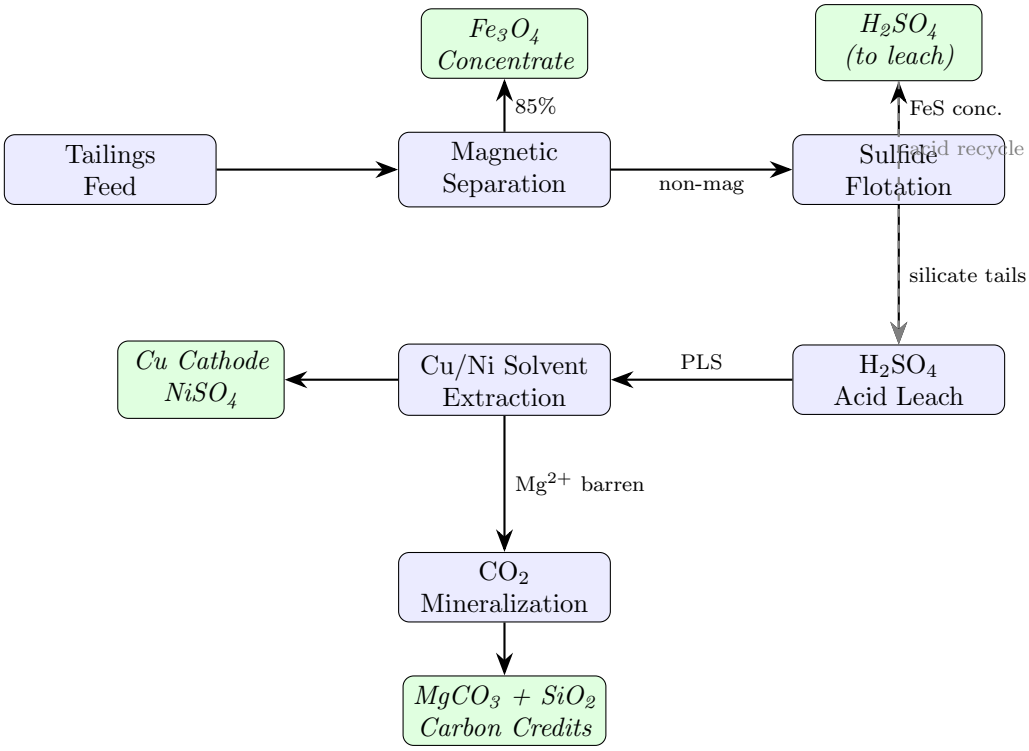

Figure 1: Process flow diagram for Eagle Mine tailings valorization.

3.1 Stage 1: Magnetic Separation

LIMS recovers magnetite at 85% recovery (IDAES separator model).

3.2 Stage 2: Sulfide Flotation

Four-stage flotation concentrates pyrrhotite at 70% recovery (IDAES flotation bank model) for H_2SO_4 production via roasting:

$$\mathrm{FeS} + \tfrac{7}{4}\mathrm{O_2} + \tfrac{1}{2}\mathrm{H_2O} \longrightarrow \tfrac{1}{2}\mathrm{Fe_2O_3} + \mathrm{H_2SO_4} \tag{1}$$

3.3 Stage 3: Acid Leach

Agitation leaching at 80 °C, 4 hours, with 80 kg H_2SO_4/t. **Reaktoro equilibrium** predicts a leach pH of **1.53**, confirming strong acid conditions for sulfide dissolution (∼87% based on empirical pH–recovery correlation).

3.4 Stage 4: Cu/Ni Solvent Extraction

McCabe–Thiele analysis at the Reaktoro-derived pH = 1.53:

- Cu: LIX 984N, $D_{\mathrm{Cu}} = 12$, O/A = 1.5, 3 stages ⇒ $R_{\mathrm{SX}} = 99.99\%$
- Ni: Cyanex 272, $D_{\mathrm{Ni}} = 8$, O/A = 1.5, 3 stages ⇒ $R_{\mathrm{SX}} = 99.95\%$

Overall recovery (dissolution × SX): $R_{\mathrm{Cu}} = 87.3\%$, $R_{\mathrm{Ni}} = 87.3\%$.

3.5 Stage 5: CO_2 Mineralization

Reaktoro Gibbs energy minimization at 185 °C, 10 bar predicts **99.99% CO_2 conversion**:

$$\mathrm{Mg^{2+}} + \mathrm{CO_2} + \mathrm{H_2O} \longrightarrow \mathrm{MgCO_3}\downarrow + 2\mathrm{H^+} \tag{2}$$

3.6 Stage 6: Thickener

Solid–liquid separation at 90% solids recovery (IDAES separator).

4 Modeling Approach

Process parameters are derived from a **three-tier** modeling hierarchy, summarized in Table 3.

Table 3: Three-tier modeling hierarchy and parameter sources.

Tier	Methodology	Parameters Derived
1. Reaktoro GEM	Gibbs energy minimization with SUPCRTBL database; aqueous speciation and mineral equilibrium	Leach pH (1.53), carbonation pH (0.87), CO_2 conversion (99.99%), Mg^{2+} speciation
2. IDAES-PSE	Sequential-modular solver with recovery-fraction and split-fraction models	Magnetic sep (85%), flotation (70%), thickener (90%)
3. McCabe–Thiele	Analytical multi-stage extraction using $R = 1 - (1 + D \cdot \mathrm{O/A})^{-n}$; dissolution informed by Reaktoro leach pH	Cu recovery (87.3%), Ni recovery (87.3%)

4.1 IDAES Sequential-Modular Solver

The flowsheet is defined as a **YAML-based DSL** comprising six unit operations connected by named streams. Units are solved in topological order:

1. Reactors dispatch to **Reaktoro** for Gibbs energy minimization (using the SUPCRTBL thermodynamic database).
2. Separators use recovery-fraction splitting.
3. SX units use McCabe–Thiele analysis with distribution-coefficient dictionaries.

4.2 Thermodynamic Backend

Reaktoro v2 with the SUPCRTBL database (Zimmer et al., 2016) provides Gibbs energy minimization. **Limitation:** Ni^{2+} and Cu^{2+} are not available in the SUPCRTBL aqueous species list. Leach dissolution is therefore estimated from an empirical pH–recovery correlation calibrated to literature data (Habashi, 1970; Free, 2013) using the *simulation-derived* leach pH of 1.53.

4.3 JAX TEA Cost Model

The `jax_tea` module implements fully differentiable proxy cost functions. The Equivalent Annualized Cost is:

$$\mathrm{EAC} = \mathrm{OPEX} + \mathrm{CAPEX} \times \frac{r}{1 - (1 + r)^{-L}} \tag{3}$$

with $r = 0.08$ and $L = 20$ years. Gradients $\nabla_{\boldsymbol{\theta}} \mathrm{EAC}$ are obtained via JAX automatic differentiation.

4.4 BoTorch Bayesian Optimization

A `SingleTaskGP` surrogate with Log Expected Improvement is used to optimize three design variables:

- Leach temperature $T \in [50, 95]\,°\mathrm{C}$
- Leach residence time $\tau \in [1, 8]\,\mathrm{h}$
- SX organic-to-aqueous ratio $\mathrm{O/A} \in [0.5, 3.0]$

Each evaluation runs the full IDAES simulation (Reaktoro + separators + SX mass balance + JAX TEA) as a black-box objective.

5 Cost Model Assumptions

Table 4: Key plant-level assumptions.

Parameter	**Value**	**Basis**
Tailings throughput	2,000 t/d	Bolt-on scale
Operating days/year	330	Mining convention
Annual tailings	660,000 t/yr	Derived
Bond Work Index	12 kWh/t	Already comminuted tails
Leach acid consumption	80 kg H_2SO_4/t	Silicate host rock
Leach temperature	80 °C	Base case (BO tested 50–95)
Leach residence time	4 hours	Base case (BO tested 1–8)
SX stages	6 (3 Cu + 3 Ni)	Counter-current
Discount rate	8%	Mining project standard
Project lifetime	20 years	Co-terminus with mine

Table 5: Commodity prices used in the revenue model.

Product	**Price**	**Source**
Cu cathode	\$8,500/t	LME 2024 avg
$NiSO_4$ equiv	\$16,000/t	LME 2024 avg
CO_2 credit	\$60/t	Voluntary market
Magnetite concentrate	\$120/t	FOB iron ore
H_2SO_4	\$80/t	Spot price

6 Results

6.1 Simulation-Derived Process Parameters

Table 6 shows the key process parameters with their source (modeling tier).

Table 6: Simulation-derived process parameters and sources.

Parameter	Value	Source
Leach pH	1.53	Reaktoro GEM
Carbonation pH	0.87	Reaktoro GEM
CO_2 conversion	99.99%	Reaktoro GEM
Mg^{2+} to carbonation	2.78%	Reaktoro mass balance
Magnetic sep recovery	85.0%	IDAES LIMS
Flotation recovery	70.0%	IDAES flotation bank
Cu dissolution (pH=1.53)	87.4%	Empirical (Reaktoro pH)
Cu SX recovery (3 stages)	99.99%	McCabe–Thiele
Cu overall recovery	**87.3%**	Dissolution × SX
Ni dissolution (pH=1.53)	87.4%	Empirical (Reaktoro pH)
Ni SX recovery (3 stages)	99.95%	McCabe–Thiele
Ni overall recovery	**87.3%**	Dissolution × SX

6.2 Capital and Operating Expenditures

Table 7: Itemized CAPEX and annual OPEX.

Process Stage	CAPEX ($M)	OPEX ($M/yr)
Tailings Handling	4.78	1.65
Regrind	1.43	0.65
Leaching	1.60	9.73
Processing (SX)	1.18	0.25
Total	**9.00**	**12.29**

6.3 Revenue

Table 8: Annual revenue from simulation-derived recoveries.

Stream	Production	Price	Revenue ($M/yr)
Ni metal	4,034 t/yr	$16,000/t	64.54
Cu metal	5,188 t/yr	$8,500/t	44.10
CO_2 credits	215,805 t/yr	$60/t	12.95
Fe_3O_4 conc.	28,050 t/yr	$120/t	3.37
H_2SO_4 surplus	0 t/yr	$80/t	0.00
Total			**124.95**

CO_2 mineralization contributes $12.95 M/yr while permanently sequestering 215,805 t CO_2/yr (Reaktoro predicts 99.99% conversion at 185 °C, 10 bar). The pyrrhotite-derived H_2SO_4 is entirely consumed by the leach circuit, providing acid self-sufficiency.

6.4 Levelized Economics

Table 9: Summary of levelized economic metrics.

Metric	Value
Equivalent Annualized Cost (EAC)	\$13.21 M/yr
Total Annual Revenue	\$124.95 M/yr
Net Annual Value	\$111.74 M/yr
Cost per tonne tailings	\$20.01/t
Revenue per tonne tailings	\$189.32/t
Net value per tonne tailings	**\$169.30/t**

6.5 BoTorch Optimization Results

Bayesian optimization (10 iterations, 5 initial LHS points) improved the net annual value from \$111.74 M/yr (base case) to **\$113.73 M/yr** (Table 10).

Table 10: BoTorch-optimized process parameters.

Design Variable	Base Case	Optimal
Leach temperature (°C)	80	50
Leach residence time (h)	4.0	4.7
SX O/A ratio	1.5	3.0
Net annual value (\$M/yr)	**111.74**	**113.73**

The optimizer favors reduced leach temperature (lower OPEX) and higher O/A ratio (improved SX extraction), gaining \$2.0 M/yr.

6.6 Cost Sensitivity Analysis

JAX automatic differentiation identifies the dominant cost drivers (Table 11).

Table 11: Top-5 cost sensitivities via JAX ∇EAC.

Parameter	ΔEAC (\$M/yr per unit)
Strip ratio	+0.990
Acid consumption (kg/t)	+0.099
Bond Work Index (kWh/t)	+0.063
Residence time (h)	+0.041
Temperature (°C)	+0.033

Acid consumption is the most operationally relevant driver: each 1 kg/t increase adds \$99 k/yr to the EAC.

7 Discussion

7.1 Modeling Fidelity

The three-tier approach provides appropriate fidelity for each process stage. Reaktoro's GEM solver is thermodynamically rigorous for the carbonation reactor (capturing the strong exergonic

driving force for $MgCO_3$ precipitation) and provides a physically meaningful leach pH that informs the downstream dissolution estimate.

The primary limitation is that Ni^{2+} and Cu^{2+} are absent from the SUPCRTBL database, preventing direct simulation of sulfide dissolution. This is addressed by coupling the Reaktoro-derived leach pH to an empirical dissolution–pH correlation from the literature, maintaining physical consistency between the acid environment and metal extraction.

7.2 Economic Drivers

The strong economics ($169/t net value) are driven by:

- High residual metal values (0.7% Ni + 0.9% Cu at LME prices)
- Modest CAPEX for a bolt-on tailings reprocessing facility
- Acid self-sufficiency from pyrrhotite roasting
- Significant CO_2 credit revenue ($12.95 M/yr)

7.3 Key Risks

- Commodity price volatility (Cu, Ni)
- Actual dissolution kinetics may differ from pH correlation
- Carbon credit price uncertainty
- Acid consumption may increase with weathered tailings

8 Conclusions

1. The IDAES + Reaktoro simulation confirms favorable conditions for all process stages, with a leach pH of 1.53 supporting high metal dissolution.
2. **CO_2 mineralization** at 185 °C shows near-complete conversion (99.99%), sequestering 215,805 t/yr and generating $12.95 M/yr in credits.
3. **Metal recovery** at 87.3% (McCabe–Thiele, informed by simulation pH) yields $108.6 M/yr combined Cu + Ni revenue.
4. BoTorch optimization improves net value by $2.0 M/yr by reducing leach temperature and increasing O/A ratio.
5. The **net value of $169/t tailings** transforms a waste stream into a major revenue center, potentially justifying reprocessing of both current and legacy stockpiles.

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